ERIC HOU

Engineering Portfolio

Hi! I'm Eric. Here's some stuff I've made.

Electrical Engineering @ Purdue University

Circuithub - Summer 2026

Making new pick-and-place head hardware for CircuitHub assembly grid



PCB + control electronics for motion and sensing, sensors and voice-coil actuators

Fusion 360 CAD and CAM

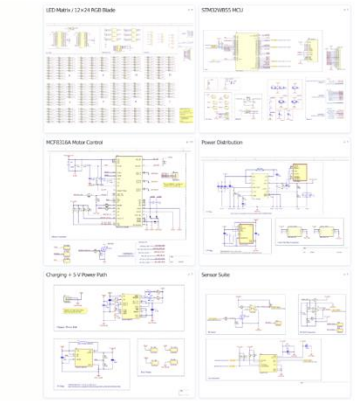
TwinCAT + Python 30,000+ motion cycles

Some details omitted for confidentiality.

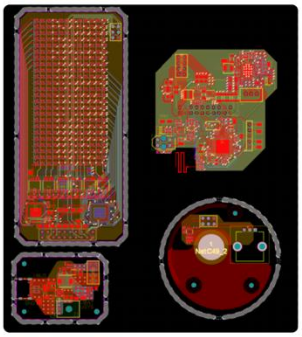
3D Volumetric Hologram Display

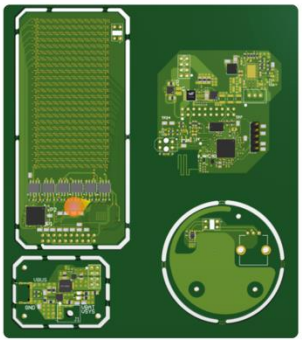
Work in progress - rotating RGB LED blade for a persistence-of-vision 3D display.

Schematic Samples



PCB Layout - In Progress





288 RGB LEDs • 1:12 multiplexing

STM32WB55 • BLE • 2.4 GHz RF

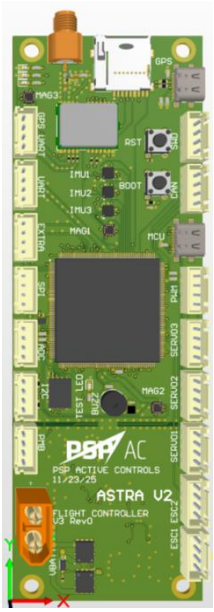
BLDC drive • motion sensing • touch input

Multiple custom boards for the stationary and rotating assemblies

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PSP Active Controls - Self-Stabilizing Rocket Lander

Circuit boards for Purdue Space Program's VTVL vehicles.



STM32H7 flight controller



Power / sensing boards

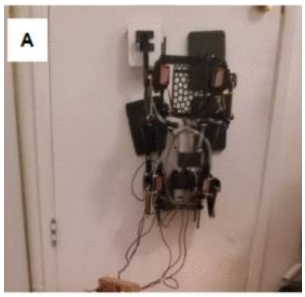


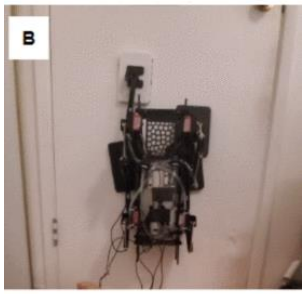
ASTRA VTVL test vehicle

STM32H7 flight controller
GPS • IMUs • radio

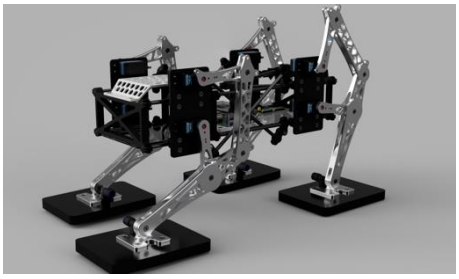
Power distribution + monitoring • CAN FD

VertiPaw - Wall Climbing Quadrupedal Robot

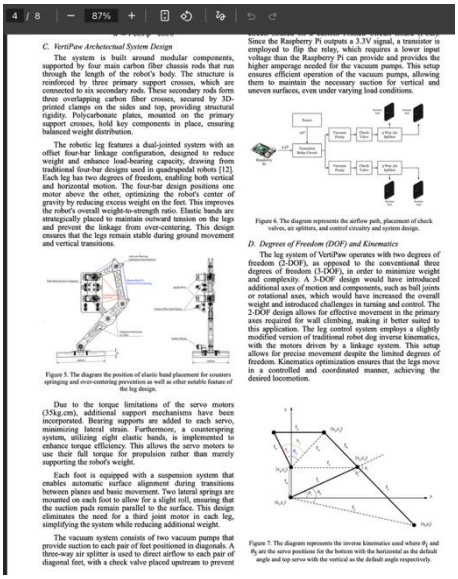








VertiPaw CAD render



4 / 8 − 87% +

C. VertiPaw Architectural System Design

The system is built around modular components, supported by four main carbon fiber chassis rods that run through the length of the robot's body. The structure is reinforced by three primary support crosses, which are connected to six secondary rods. These secondary rods form three overlapping carbon fiber crosses, secured by 3D-printed clamps on the sides and top, providing structural rigidity. Polycarbonate plates, mounted on the primary support crosses, hold key components in place, ensuring balanced weight distribution.

The robotic leg features a dual-jointed system with an offset four-bar linkage configuration, designed to reduce weight and enhance load-bearing capacity, drawing from traditional four-bar designs used in quadrupedal robots [12]. Each leg has two degrees of freedom, enabling both vertical and horizontal motion. The four-bar design positions one motor above the other, optimizing the robot's center of gravity by reducing excess weight on the feet. This improves the robot's overall weight-to-strength ratio. Elastic bands are strategically placed to maintain outward tension on the legs and prevent the linkage from over-centering. This design ensures that the legs remain stable during ground movement and vertical transitions.

Figure 5. The diagram the position of elastic band placement for counters springing and over-centering prevention as well as other notable feature of the leg design.

Due to the torque limitations of the servo motors (35kg.cm), additional support mechanisms have been incorporated. Bearing supports are added to each servo, minimizing lateral strain. Furthermore, a counterspring system, utilizing eight elastic bands, is implemented to enhance torque efficiency. This allows the servo motors to use their full torque for propulsion rather than merely supporting the robot's weight.

Each foot is equipped with a suspension system that enables automatic surface alignment during transitions between planes and basic movement. Two lateral springs are mounted on each foot to allow for a slight roll, ensuring that the suction pads remain parallel to the surface. This design eliminates the need for a third joint motor in each leg, simplifying the system while reducing additional weight.

The vacuum system consists of two vacuum pumps that provide suction to each pair of feet positioned in diagonals. A three-way air splitter is used to direct airflow to each pair of diagonal feet, with a check valve placed upstream to prevent

Since the Raspberry Pi outputs a 3.3V signal, a transistor is employed to flip the relay, which requires a lower input voltage than the Raspberry Pi can provide and provides the higher amperage needed for the vacuum pumps. This setup ensures efficient operation of the vacuum pumps, allowing them to maintain the necessary suction for vertical and uneven surfaces, even under varying load conditions.

Figure 6. The diagram represents the airflow path, placement of check valves, air splitters, and control circuitry and system design.

D. Degrees of Freedom (DOF) and Kinematics

The leg system of VertiPaw operates with two degrees of freedom (2-DOF), as opposed to the conventional three degrees of freedom (3-DOF), in order to minimize weight and complexity. A 3-DOF design would have introduced additional axes of motion and components, such as ball joints or rotational axes, which would have increased the overall weight and introduced challenges in turning and control. The 2-DOF design allows for effective movement in the primary axes required for wall climbing, making it better suited to this application. The leg control system employs a slightly modified version of traditional robot dog inverse kinematics, with the motors driven by a linkage system. This setup allows for precise movement despite the limited degrees of freedom. Kinematics optimization ensures that the legs move in a controlled and coordinated manner, achieving the desired locomotion.

Figure 7. The diagram represents the inverse kinematics used where θ_1 and θ_2 are the servo positions for the bottom with the horizontal as the default angle and top servo with the vertical as the default angle respectively.

Vacuum adhesion • 9 lb payload

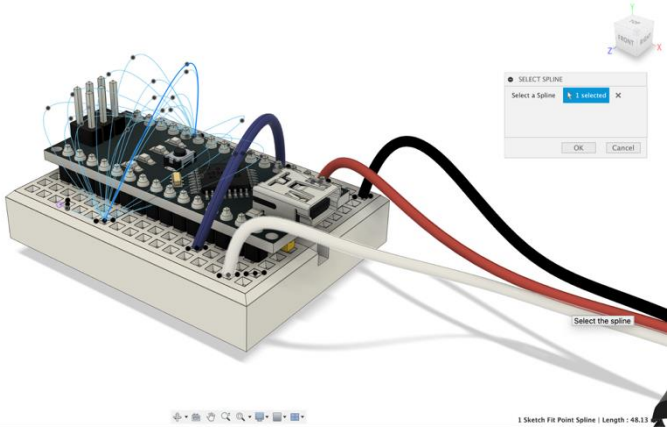
LiDAR localization + inverse-kinematics path planning

Carbon fiber / polycarbonate / 3D printed structure

Paper published in IEEE IRC 2024
First author

Fusion 360 Wire Generator

Parametric wire-routing plug-in.



5,000+ users • 5★ on Autodesk

Python + Fusion 360 API

Parameterized 3D spline routing

Vector math + geometric constraints

5-Stage Electromagnetic Coil Launcher

Custom switching electronics + mechanical assembly.





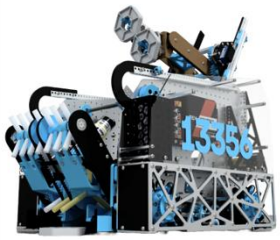


43 A measured discharge pulses

MOSFET switching PCBs for supercapacitor bank

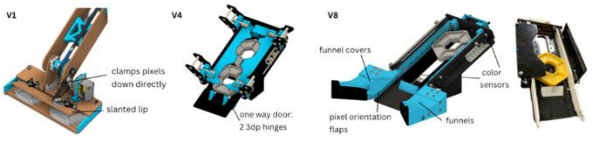
FIRST Robotics

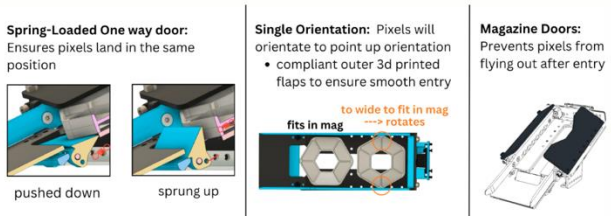
Former Hardware Lead • 2021–2025











#3 California • Top 30 worldwide • MTI 3× (Top 40 / 7000 teams)

Four years of end-to-end robot hardware
CAD, fabrication, electronics, controls, and validation.

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